

Explanatory Material

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This explanatory material aims to: document the data processing pipeline, present the main results of the exploratory analysis, and describe the modeling approach applied.

1 General Information

The dataset used is publicly available on GitHub [5].

The dataset contains **9,994 rows** and **20 columns**, listed as follows: Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country, City, State, Postal Code, Region, Product ID, Category, Sub-Category, Product Name, Sales, Quantity, Discount, and Profit.

This is a dataset from a fictional US retail company called **Superstore**, specializing in furniture, office supplies, and technology products. The dataset provides a detailed analysis of the company's performance from 2019 to 2022.

The compiled work is structured as follows:

```
.
|-- data
|   |-- data_processed
|   '-- data_raw
|-- main.py
|-- model
|   '-- SUPERSTORE_CLUSTERING.csv
|-- notebooks
|   |-- 01_initial_exploration.ipynb
|   |-- 02_outliers.ipynb
|   |-- 03_kmeans.ipynb
|   '-- 04_temporal_series.ipynb
|-- output
|   '-- models_predict
|       |-- arima_cluster_0.joblib
|       |-- arima_cluster_1.joblib
|       |-- arima_cluster_2.joblib
|       '-- kmeans_clustering.joblib
|-- README.md
|-- report
|   |-- notebook_01
|   |-- notebook_03
|   '-- notebook_04
|-- requirements.txt
'-- src
    |-- build_model.py
    |-- predict.py
    '-- utils.py
```

where:

- **data/**: Stores the project data.
 - **data_raw/**: Contains the raw, unprocessed data.
 - **data_processed/**: Contains the cleaned data, ready for modeling.
- **model/**: Contains intermediate modeling artifacts, such as datasets with assigned clusters.
- **notebooks/**: Gathers analysis notebooks and experiments (EDA, outlier handling, clustering, and time series).

- `output/`: Stores the final model outputs.
 - `models_predict/`: Contains serialized trained models (`.joblib` files).
- `report/`: Directory for storing report artifacts such as figures, charts, and tables generated throughout the analyses.
- `src/`: Main source code of the project.
 - `build_model.py`: Responsible for training the models.
 - `predict.py`: Responsible for generating predictions.
 - `utils.py`: Auxiliary functions used throughout the project.
- `main.py`: Main script that runs the project pipeline.
- `README.md`: General project documentation.
- `requirements.txt`: List of project dependencies.

2 Data Processing

To process the dataset and generate the file `SUPERSTORE_PROCESSED.csv`, we followed these steps:

1. Standardization of string columns;
2. Removal of irrelevant columns (`CUSTOMER_NAME`, `POSTAL_CODE`, and `ORDER_ID`);
3. Processing of date columns;
4. Removal of inconsistent and duplicate records;
5. Treatment of location columns;
6. Creation of the `NET_SALES` column using the formula below:

$$\text{net_sales} = \text{sales} \times \text{quantity} \times (1 - \text{discount})$$

and removal of the `SALES`, `QUANTITY`, and `DISCOUNT` columns.

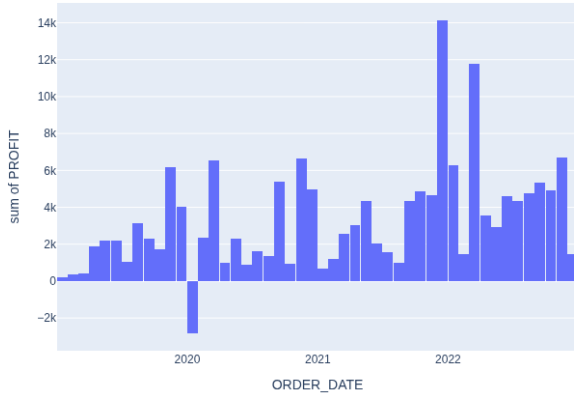
Through this data processing pipeline, we obtained a dataset that retains substantial information and is ready for exploratory analyses both in a Python environment and in platforms such as Excel, Google Sheets, Power BI, and Google Looker. Finally, it is worth noting that the data processing code is available in `utils.py`.

3 Exploratory Analysis

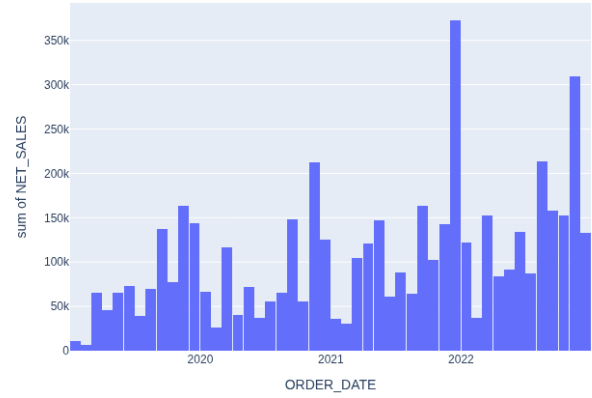
Disclaimer: We predominantly used the *plotly* library [3], which stands out for generating interactive charts. This feature allows for a deeper exploration of the data when viewed in a Python environment. Although the main insights are presented in this document, we recommend consulting the notebook `01_initial_exploration.ipynb` for a more complete and interactive analysis.

3.1 Evolution of Net Sales and Profit

Below are two charts showing, respectively, the evolution of net sales (`NET_SALES`) and profit over time. In general, both show growth throughout 2019, followed by fluctuations in subsequent years, with occasional peaks in both sales and profit.



(a) Profit distribution



(b) Net sales distribution

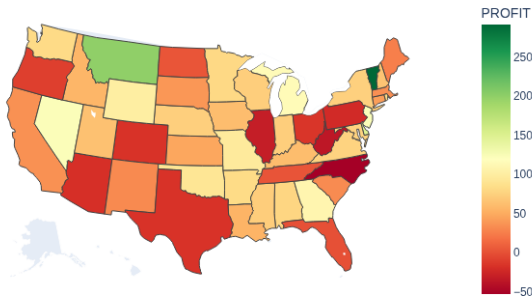
Figure 1: Profit and sales distributions.

3.2 Profit by Geographic Location

Regarding sales by geographic location, New York and California are the states with the highest total profit volume, at US\$ 41,597.96 and US\$ 35,428.39, respectively, while the worst performers are Illinois at US\$ -8,785.98 and Texas at US\$ -8,517.69.

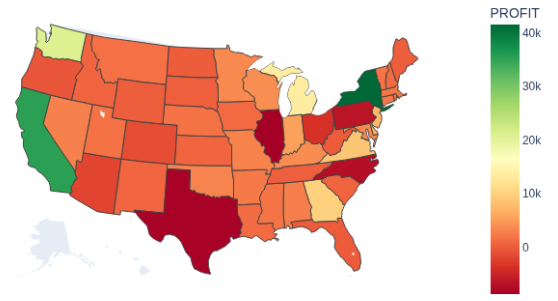
The states with the best average profit are Vermont (US\$ 291.83) and Montana (US\$ 200.90), while the worst are West Virginia (US\$ -36.94) and Illinois (US\$ -31.83).

Average Profit by State in the USA



(a) Average profit by state

Total Profit by State in the USA



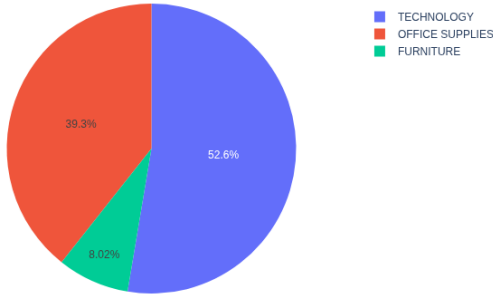
(b) Total profit by state

Figure 2: Geographic distribution of profit.

3.3 Distribution Charts

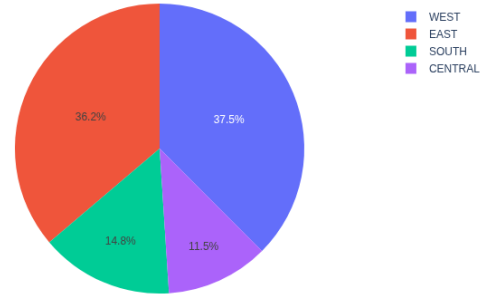
- **Profit by category:** Technology accounts for the largest share of profit (52.6%), followed by Office Supplies (39.3%), with Furniture contributing only 8.0%.
- **Profit by region:** The West (37.5%) and East (36.2%) regions lead in profit, while South (14.8%) and Central (11.5%) contribute less.
- **Shipping mode:** Standard Class is dominant (59.8%), followed by Second Class (19.2%) and First Class (15.7%), with Same Day rarely used (5.3%).

Profit by Category



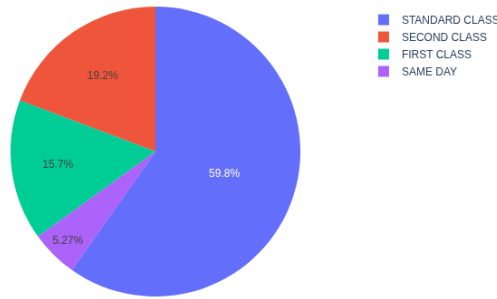
(a) Profit distribution by category.

Profit by Region



(b) Profit distribution by region.

Most Frequent Ship Mode (by number of orders)



(c) Shipping mode frequency.

Figure 3: Distribution charts.

4 Modeling

4.1 Clustering

The goal of the clustering process is to profile the groups of customers who shop at Superstore and to understand the consumption pattern of each group. For a more precise analysis, it is necessary to group customers according to their consumption patterns.

4.1.1 Preprocessing

Starting from the processed dataset `SUPERSTORE_PROCESSED.csv`, we performed the following preparation for the clustering process:

1. Selection of relevant columns for clustering (`ORDER_DATE`, `SHIP_MODE`, `CUSTOMER_ID`, `SEGMENT`, `CATEGORY`, `REGION`, `NET_SALES`, `PROFIT`, `NET_SALES_SCALED`, and `PROFIT_SCALED`);
2. Aggregation by customer;
3. Grouping by customer and aggregation of relevant columns;
4. Data preparation for clustering (one-hot encoding).

As with the previous processing step, the model base preprocessing is available in `utils.py`.

4.1.2 Model Parameter Selection

The model used for the clustering process was K-Means, available in the Scikit-learn library [2]. To define the number of clusters, we analyzed the following metrics: Inertia, Silhouette Score, Calinski-Harabasz Index, and Davies-Bouldin Index. The best values of k for each metric are defined in Table 1.

Table 1: Selection of number of clusters by different metrics.

Metric	Number of Clusters (k)
Silhouette Score	3
Calinski-Harabasz	2
Davies-Bouldin	7
Elbow Method	—

In addition, we obtained the following charts:

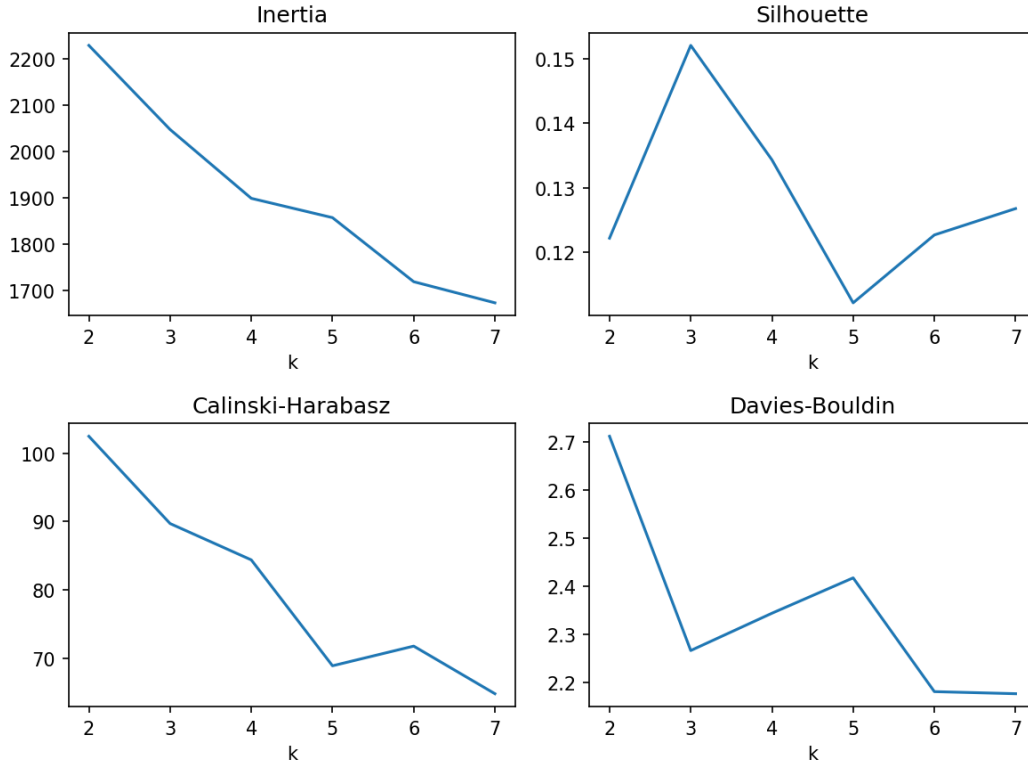


Figure 4: Analysis of different metrics to define the number of clusters.

Note in the chart that, despite the different values in Table 1, the value that delivers the best overall results is $k = 3$.

4.1.3 Model Application

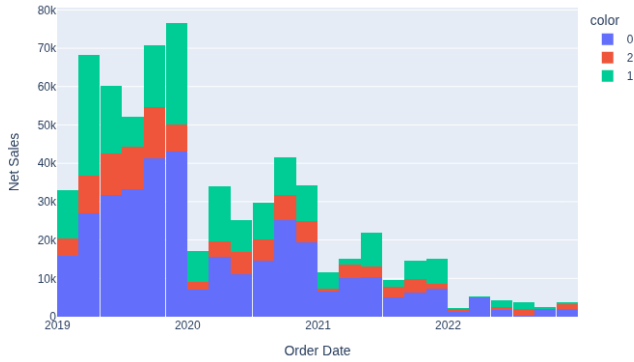
Applying the model, we can observe the following average profit (PROFIT) and net sales (NET_SALES) for each group:

Table 2: Average statistics by cluster.

Cluster	No. Customers	Avg. Net Sales	Avg. Profit
0	395	868.46	56.02
1	229	889.33	56.64
2	146	722.67	49.42

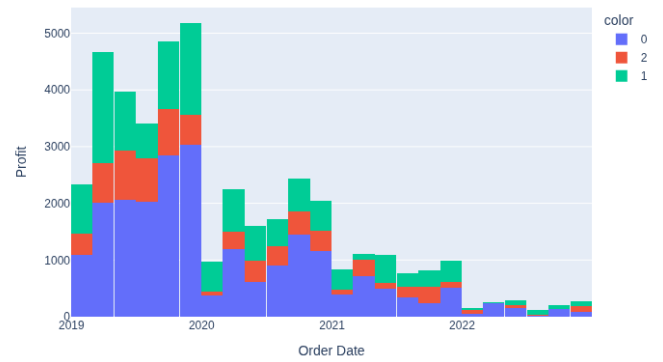
Graphically, we obtain:

Net Sales Histogram by Cluster Over Time



(a) Net sales histogram by cluster over time.

Profit Histogram by Cluster Over Time



(b) Profit histogram by cluster over time.

Figure 5: Net sales and profit histograms by cluster over time.

4.2 Prediction

Based on the clusters defined in the previous section, we will build a prediction model using time series analysis, where the target variable is the `PROFIT` column. Our objective is to forecast profit for the coming months based on the available historical data.

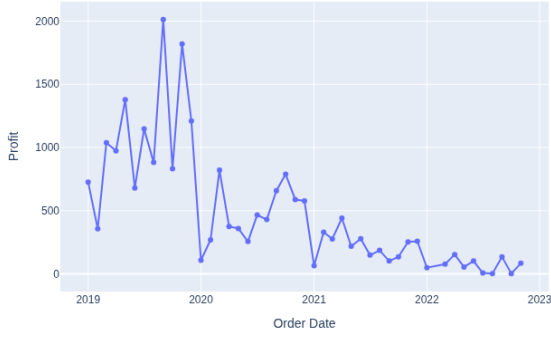
4.2.1 Preprocessing

We followed the procedure below to prepare the data for the prediction model:

1. Preparation of data for temporal analysis by cluster;
2. Conversion of the `ORDER_DATE` column to the `datetime` type;
3. Grouping by month and cluster, aggregating total profit;
4. Chronological sorting and filtering of data to consider only observations from July 2020 onward;
5. Splitting the data into subsets by cluster (0, 1, and 2) and storing each subset in a dictionary indexed by cluster;
6. Application of the stationarity test and series adjustment (via differencing, if necessary).

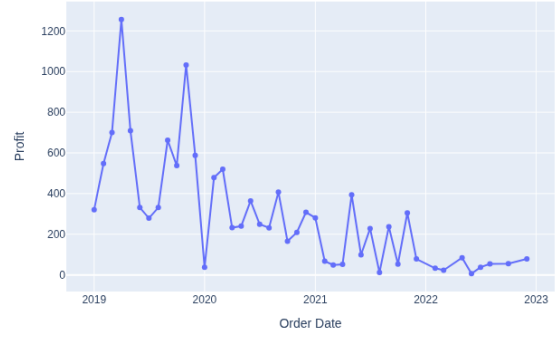
A **fundamental** step for the model to perform well was filtering the data to include only observations from July 2020 onward. Observe the following charts:

Monthly Profit - Cluster 0



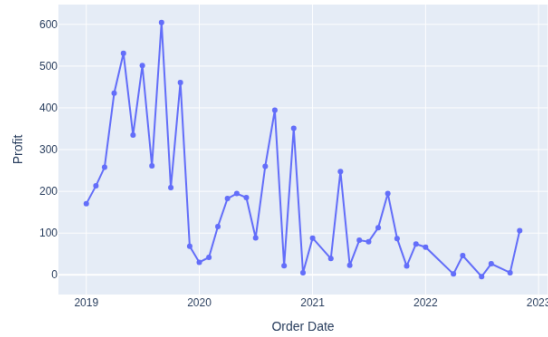
(a) Monthly profit evolution for Cluster 0.

Monthly Profit - Cluster 1



(b) Monthly profit evolution for Cluster 1.

Monthly Profit - Cluster 2



(c) Monthly profit evolution for Cluster 2.

Figure 6: Monthly profit evolution by cluster.

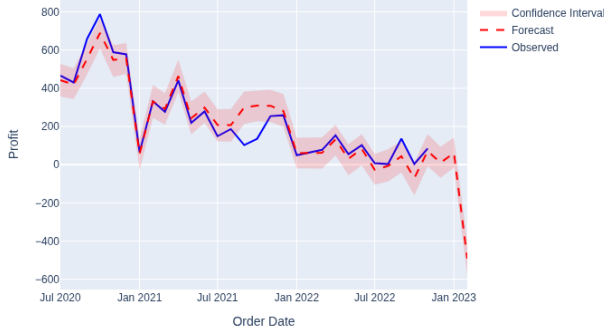
Note that there is a major shift in the consumption pattern across all clusters from 2020 onward. Therefore, we chose to exclude data prior to the second half of 2020 in order to feed the model a consumption pattern closer to the period we wish to forecast.

4.2.2 Model Application

For modeling, we used the automatic ARIMA (*AutoRegressive Integrated Moving Average*) model available in the *pmdarima* library [4], as we are dealing with a time series. The use of this model offers the advantage of automatically determining the ideal parameter values, improving model performance [1]. The results are shown below:

As with previous processing steps, the procedure is saved in `utils.py`.

Profit Forecast - Cluster 0 (3 months) | $R^2 = 0.905$ | MAE = -0.097 | MSE = 4246.375



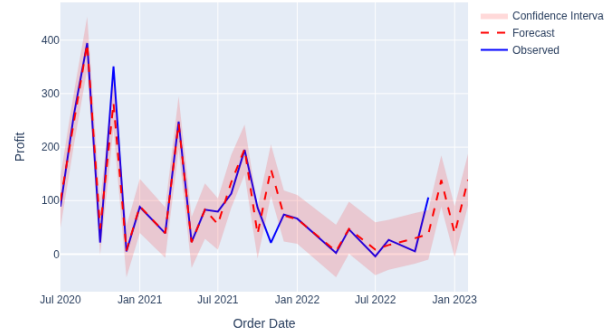
(a) Monthly profit forecast for Cluster 0 with a 3-month horizon.

Profit Forecast - Cluster 1 (3 months) | $R^2 = 0.927$ | MAE = 0.083 | MSE = 1068.026



(b) Monthly profit forecast for Cluster 1 with a 3-month horizon.

Profit Forecast - Cluster 2 (3 months) | $R^2 = 0.879$ | MAE = -0.078 | MSE = 1403.030



(c) Monthly profit forecast for Cluster 2 with a 3-month horizon.

Figure 7: Monthly profit forecast by cluster with a 3-month horizon.

4.3 Conclusion and Perspectives

This work encompassed the processing and preparation of the dataset, exploratory analysis to identify sales and profit patterns, customer segmentation through clustering, and finally, time series modeling to forecast profit by group. In this way, it was possible not only to understand the behavior of the data, but also to obtain a consistent foundation to support decision-making.

References

- [1] Aditi Babu. A guide to time series forecasting in python. <https://medium.com/@aditib259/a-guide-to-time-series-forecasting-in-python-b0b4529045db>, 2023. Accessed 22 mar. 2026.
- [2] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [3] PlotlyTechnologiesInc. Collaborative data science, 2015.
- [4] Taylor G. Smith et al. pmdarima: Arima estimators for Python, 2017–. [Online; accessed 2026-03-22].
- [5] Candice Wu. Superstore sales analysis dataset. <https://raw.githubusercontent.com/WuCandice/Superstore-Sales-Analysis/refs/heads/main/dataset/Superstore%20Dataset.csv>, n.d. Accessed: 2026-03-20.